

INDUSTRIAL PROCESS MODELLING & SIMULATION TRAINING

8 – 10 August 2022

Part 3 : Distillation Modelling & Simulation

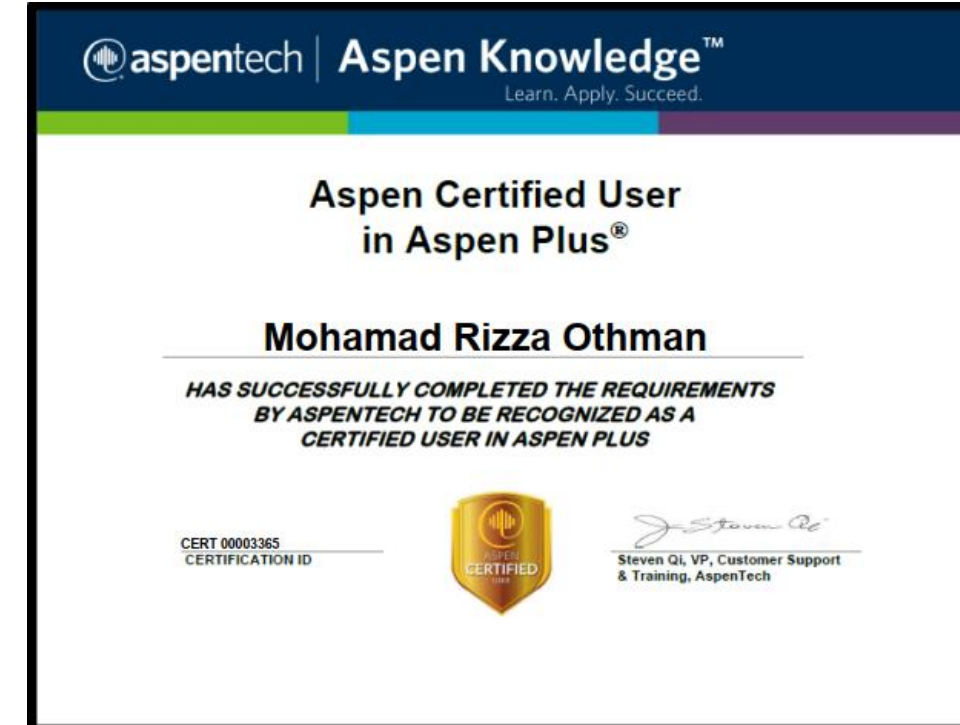
Trainer Background

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- Education background
 - B. Eng. (Chem. Eng.) – UTM 2002
 - M. Eng. (Chem. Eng.) – UTM 2005
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 - PEng or Ir. (BEM)
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Aspen Plus Background

- More than 15 years experience in Aspen Plus, Dynamics, ASW, EDR.
- Taught UG/PG courses i.e., Process control & dynamics, process simulation & advanced Process Simulation.
- Research on modelling biodiesel, acetic acid, hydrogen production, oleochemical, distillation, extraction, DWC, MSR, CO2 absorption etc.
- Certified Aspen Plus User
- Presenter at OPTIMIZE '21 & BEYOND OPTIMIZE '21.



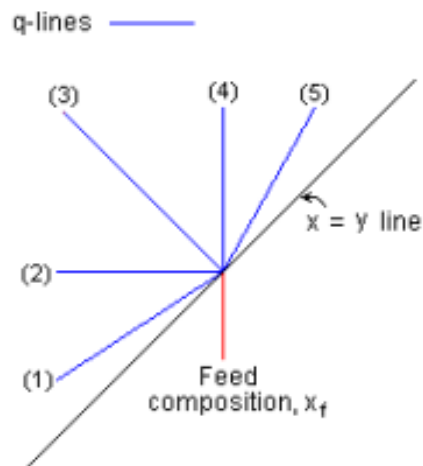
Topics

- 3.1 Distillation modelling using shortcut & rigorous model
- 3.2 Model analysis & design specifications
- 3.3 Utilities setup and analysis
- 3.4 Column internals

Distillation Modelling

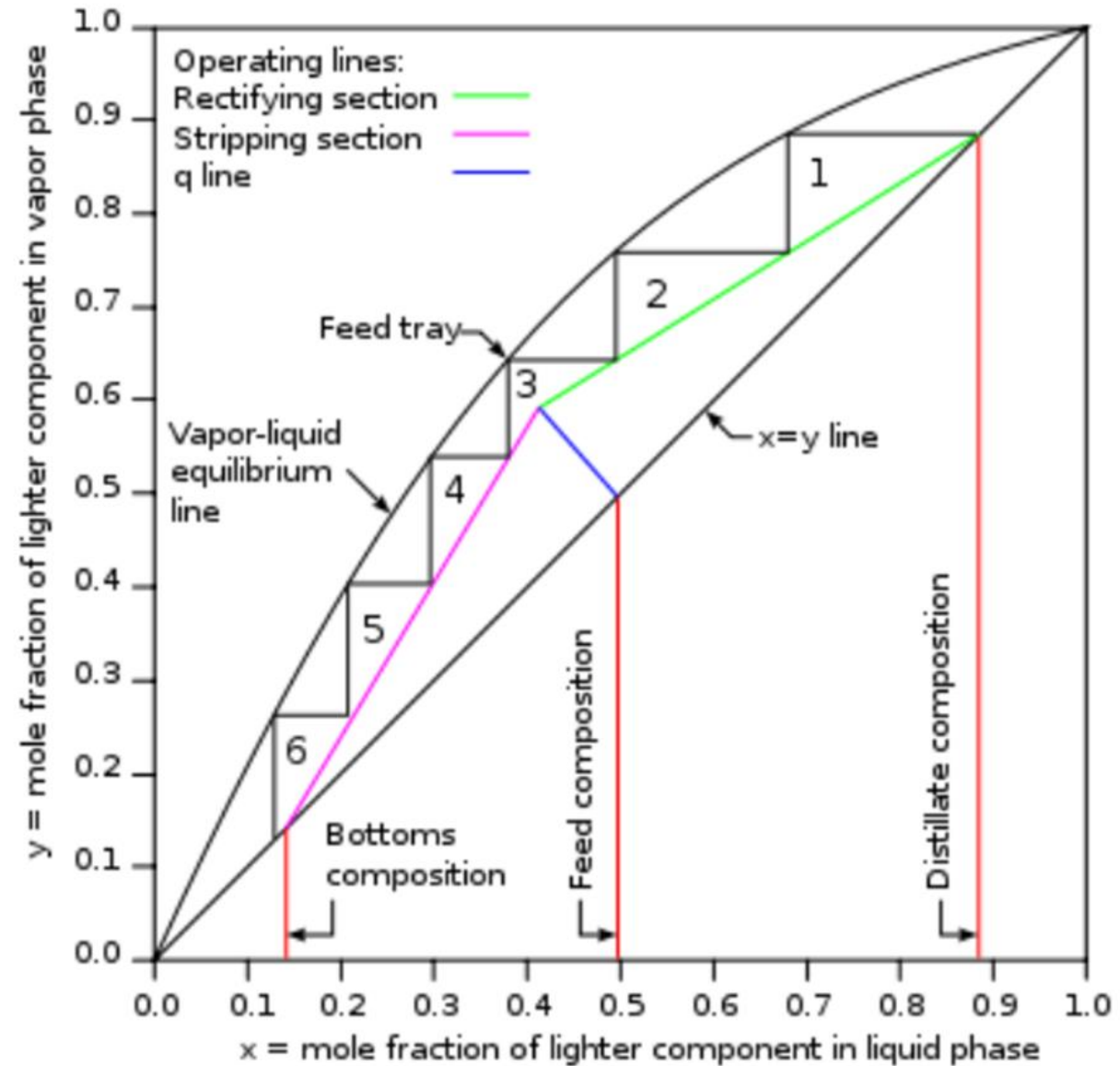
Binary Components

- Involve 2 components.
- For analysis of binary component using McCabe-Thiele method to find number of stages.



Example q-line slopes:

- (1) Superheated vapor feed
- (2) Saturated vapor feed
- (3) Partially vaporized feed
- (4) Saturated liquid feed
- (5) Subcooled liquid feed



Multicomponent

- Involve more than 2 components.
- Light key (LK) is the least volatile component enriched in the distillate stream.
- Heavy key (HK) is the most volatile component enriched in the bottoms stream.
- Example, consider component A, B, C, and D with decreasing volatility. A the most volatile, D the least volatile.

CASE	1	2	3
Distillate	A	AB	ABC
Bottom	BCD	CD	D
LK			
HK			

Shortcut Method : DSTWU & DISTL

- Used for initial design stage.
- Requires much less information and specifications.
- Assumes (i) constant relative volatilities and (ii) constant molar overflow.
- 2 shortcut model in Aspen Plus i.e., DSTWU & DISTL.

Shortcut Method : DSTWU & DISTL

- DSTWU uses,
 - Winn equation to estimate N, Underwood equation to calculate minimum RR and Gilliland equation to calculate the required RR at a specified N or vice versa.
- DISTL uses,
 - Edmister's method to calculate column compositions for a given number of N and RR.
- DSTWU provides a very good starting point for distillation column design but not suitable for strongly non-ideal mixtures (e.g., ethanol-water).

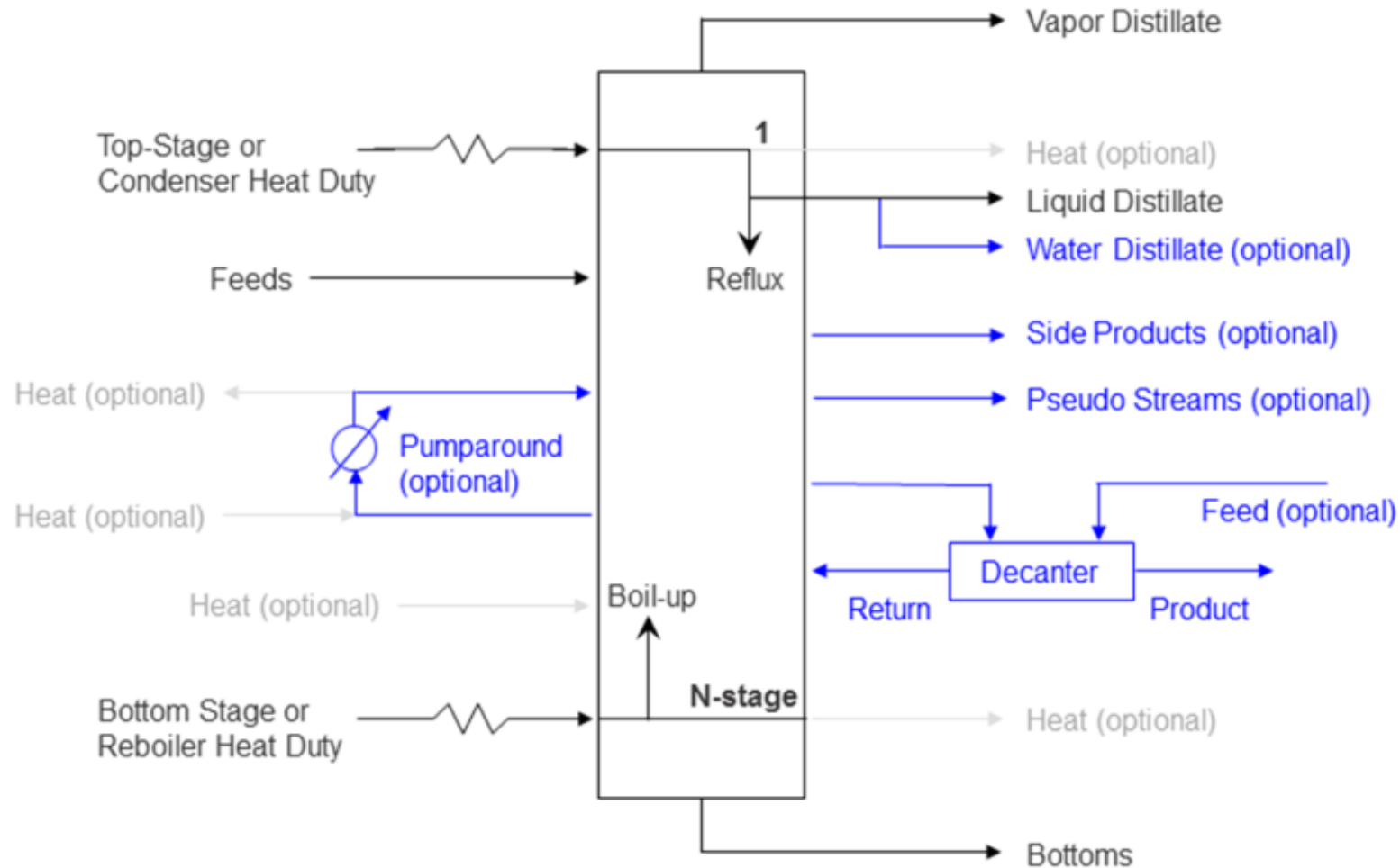
Rigorous Method : RadFrac

- Distillation model block for rigorous calculations.
- Suitable for highly non-ideal liquid solutions
- Processes with an on-going chemical reaction.
- Able to have multiple feed and product streams (including pump-around streams).
- Able to perform hydraulic analysis with trays, random packing, or structured packing.

Rigorous Method : RadFrac

MODEL	DESCRIPTION	PURPOSE	USE
RadFrac	Rigorous fractionation	Rigorous rating and design for single columns	Distillation, absorbers, strippers, extractive and azeotropic distillation, reactive distillation
MultiFrac	Rigorous fractionation for complex columns	Rigorous rating and design for multiple columns of any complexity	Heat integrated columns, air separators, absorber/stripper combinations, ethylene primary fractionator/quench tower combinations, petroleum refining
PetroFrac	Petroleum refining fractionation	Rigorous rating and design for petroleum refining applications	Preflash tower, atmospheric crude unit, vacuum unit, catalytic cracker or coker fractionator, vacuum lube fractionator, ethylene fractionator and quench towers
BatchFrac*+	Rigorous batch distillation	Rigorous rating calculations for single batch columns	Ordinary azeotropic batch distillation, 3-phase, and reactive batch distillation
RateFrac*.	Rate-based distillation	Rigorous rating and design for single and multiple columns. Based on nonequilibrium calculations	Distillation columns, absorbers, strippers, reactive systems, heat integrated units, petroleum applications
Extract	Liquid-liquid extraction	Rigorous rating for liquid-liquid extraction columns	Liquid-liquid extraction

RadFrac Flowsheet Connectivity



Input Specifications: Configuration

1. **Calculation type:** Equilibrium, Rate-based
2. **Number of stages**
3. **Condenser type:** Total, partial, partial V-L, none
4. **Reboiler:** Kettle. Thermosiphon, none
5. **Valid phases:** VL, VLL, etc.
6. **Convergence:** Standard, petroleum/ide boiling, strong ideally non ideal liquid, azotropic, cyrogenic, custom
7. **2 operating specifications:** distillate rate, bottom rate, reflux rate, boilup rate, boilup ratio, distillate to feed ratio, bottoms to feed ratio, condenser duty, reboiler duty.

Main Flowsheet x COLUMN (RadFrac) x +

Configuration Streams Pressure Condenser Reboiler 3-Phase Comments

Setup options

Calculation type: *Equilibrium*

Number of stages: 50 Stage Wizard

Condenser: *Partial-Vapor*

Reboiler: *Kettle*

Valid phases: *Vapor-Liquid*

Convergence: *Strongly non-ideal liquid*

Operating specifications

Distillate rate: *Mass* 50 kg/hr

Reflux ratio: *Mole* 5

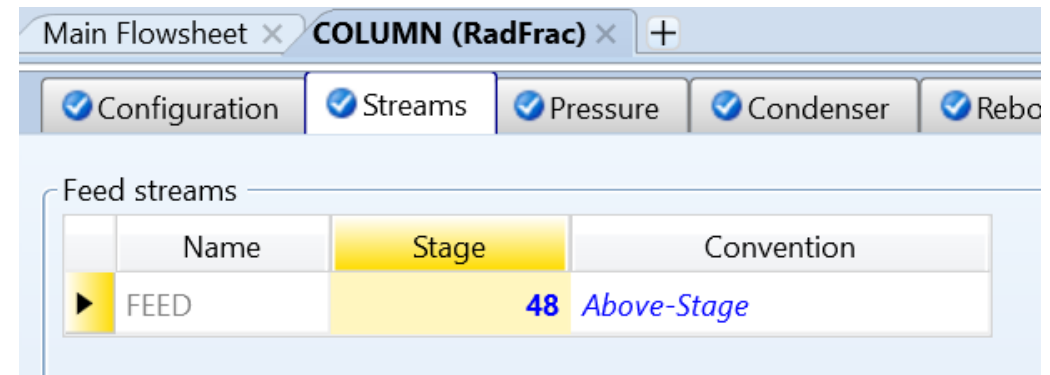
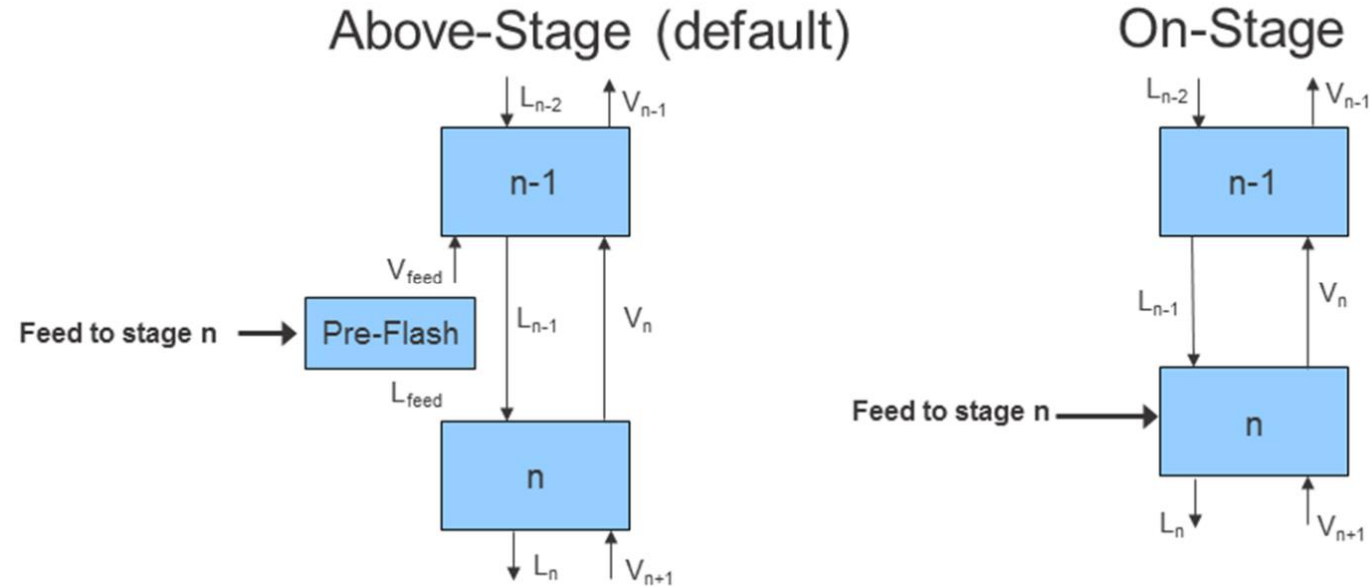
Free water reflux ratio: 0 Feed Basis

Design and specify column internals

Input Specifications: Streams

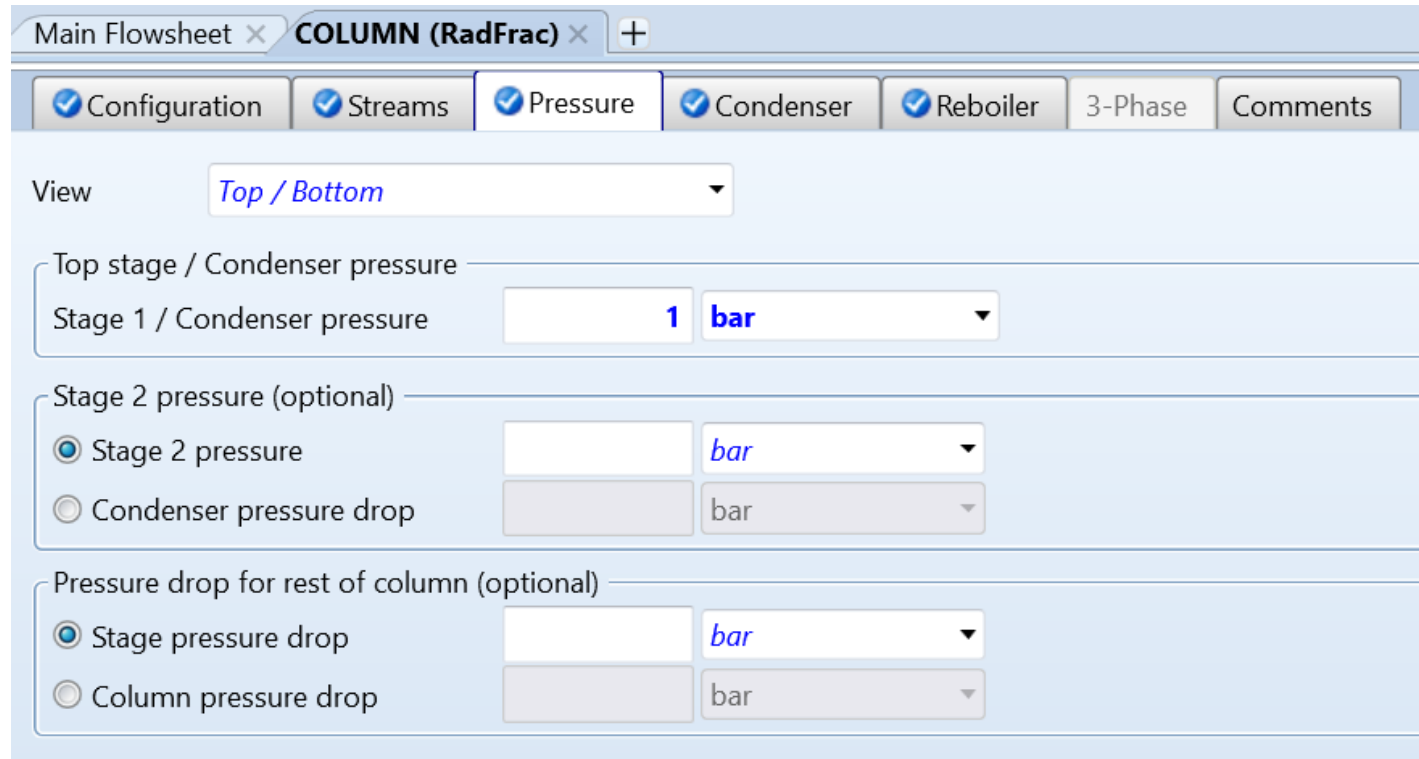
Feed streams:

- ABOVE-STAGE: Vapor from feed goes to stage above feed stage, Liquid goes to feed stage.
- ON-STAGE / LIQUID / VAPOR: Vapor and / or liquid from feed go to specified feed stage.
- For most realistic column scenarios, Above-Stage should be used.



Input Specifications: Streams

- **Pressure,**
 - Column pressure profile
 - Top/Bottom pressure
 - Section pressure drop

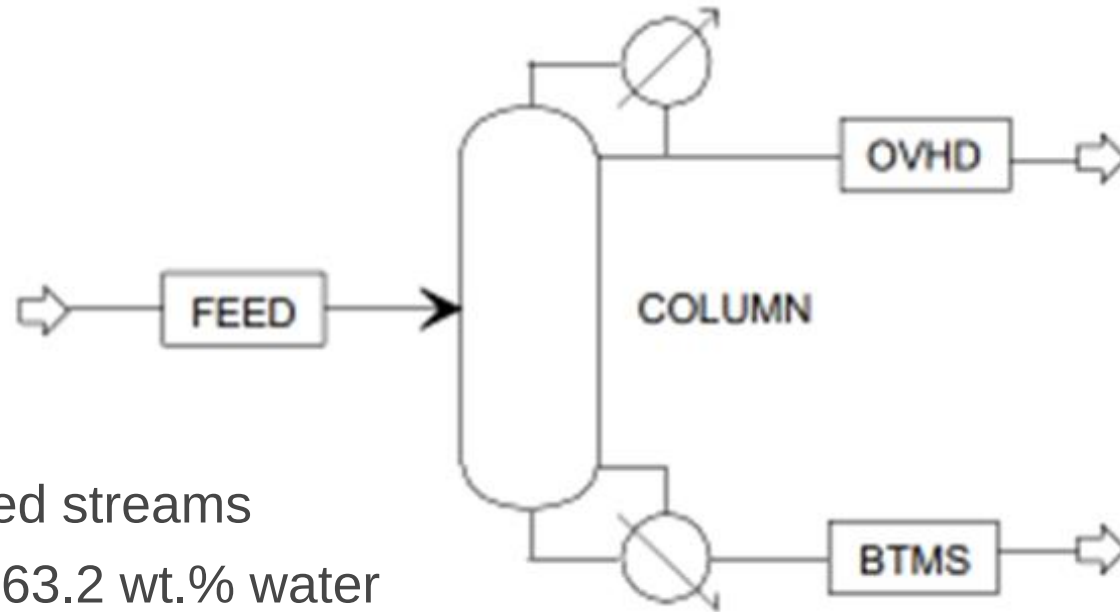


The screenshot shows the 'COLUMN (RadFrac)' configuration window with the 'Pressure' tab selected. The window has a tab bar at the top with 'Main Flowsheet', 'COLUMN (RadFrac)', and a '+' icon. Below the tab bar are several sub-tabs: 'Configuration', 'Streams', 'Pressure' (selected), 'Condenser', 'Reboiler', '3-Phase', and 'Comments'. The 'View' dropdown is set to 'Top / Bottom'. The 'Top stage / Condenser pressure' section has a text box for 'Stage 1 / Condenser pressure' with the value '1' and a unit dropdown set to 'bar'. The 'Stage 2 pressure (optional)' section has two radio buttons: 'Stage 2 pressure' (selected) and 'Condenser pressure drop'. The 'Stage 2 pressure' option has a text box and a unit dropdown set to 'bar'. The 'Condenser pressure drop' option has a text box and a unit dropdown set to 'bar'. The 'Pressure drop for rest of column (optional)' section has two radio buttons: 'Stage pressure drop' (selected) and 'Column pressure drop'. The 'Stage pressure drop' option has a text box and a unit dropdown set to 'bar'. The 'Column pressure drop' option has a text box and a unit dropdown set to 'bar'.

Example 3.1

A saturated vapor stream containing 68.5 wt.% ethylene and 31.5 wt.% ethane at 25 bar with a total flowrate of 7.3 million kg/day must be separated. Report a reasonable design parameters estimation using **DSTWU** model block required to achieve a separation of 99.9% recovery of ethylene and 99.0% recovery of ethane. Use Peng-Rob thermodynamic model.

Example 3.2 : DC Using RADFRAC



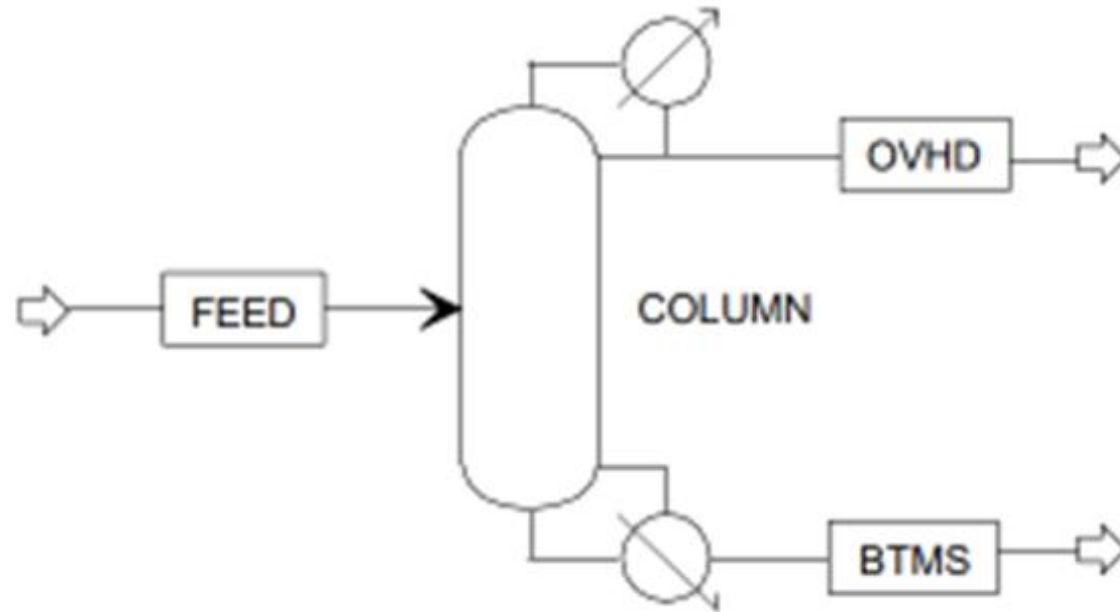
Feed streams

- 63.2 wt.% water
- 36.8 wt.% methanol
- Total flow: 120,000 lb/hr
- Pressure: 18 psia
- Saturated liquid
- NRTL-RK t/dynamic method

Column specifications

- 38 trays (40 stages)
- Feed tray: 23 (stage 24)
- Total condenser
- Top stage $P = 16.1$ psia
- Distillate flowrate: 1245 lbmol/hr
- Molar reflux ratio: 0.5

Example 3.2a : DC Using RADFRAC

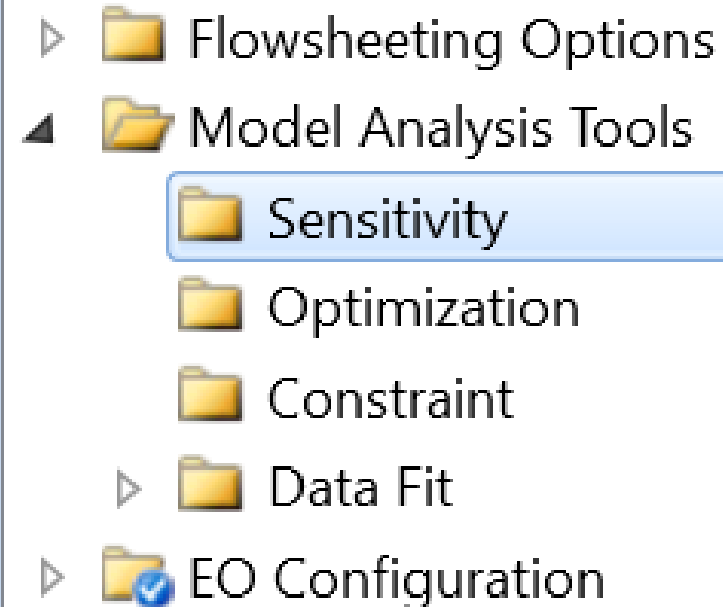


For the same problem, simulate the process with pressure drop per stage of 0.1 psi and report your results.

Model Analysis & Design Specifications

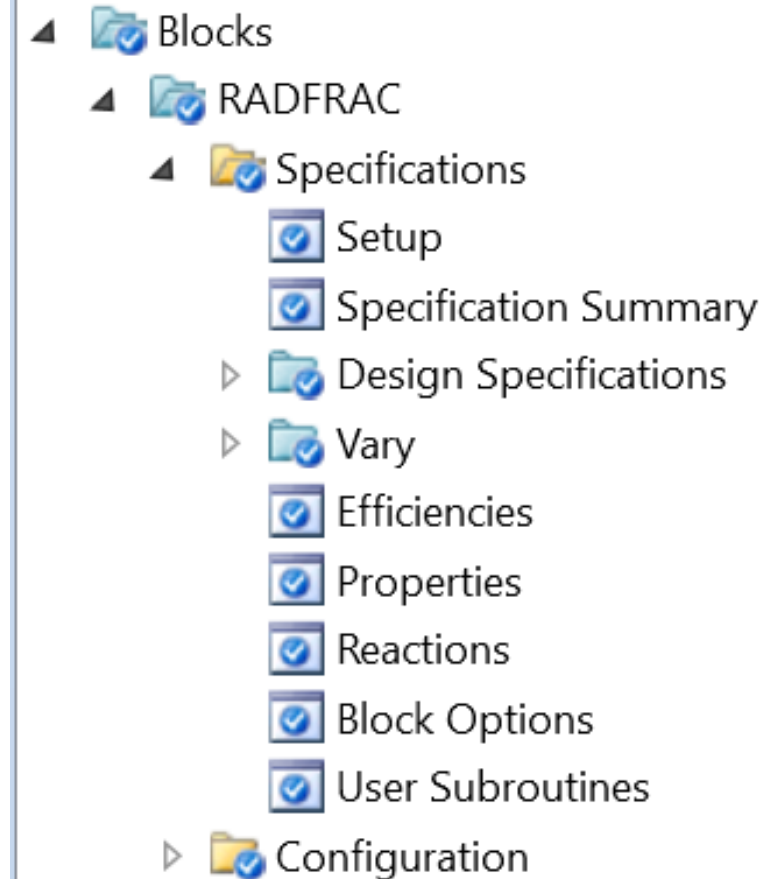
Sensitivity Analysis (SA)

- Provides mechanism for analyzing flowsheet behavior.
- Need to identify manipulated variable (MV), range and output or sampled variable (SV).
- MV is the input values and SV is calculated value.
- Allow multiple MVs and SVs but will increase the iterations.
- Can be used for other calculations i.e., design spec, optimization



Design Specifications (DS)

- DS sets the value of a variable that Aspen Plus would otherwise calculate.
- For each DS, need to identify a design variable (DV) in SPECIFICATIONS FORM and manipulated variable (MV) in VARY FORM to meet the specification.
- DS achieves its objective by manipulating an input variable specified by the user.
- Quantities that are calculated during the simulation should not be varied directly. For example, the stream flow rate of a recycle stream cannot be varied; however, the split fraction of an FSplit block where the recycle stream is an outlet can be varied.
- A design specification can only manipulate the value of one input variable.

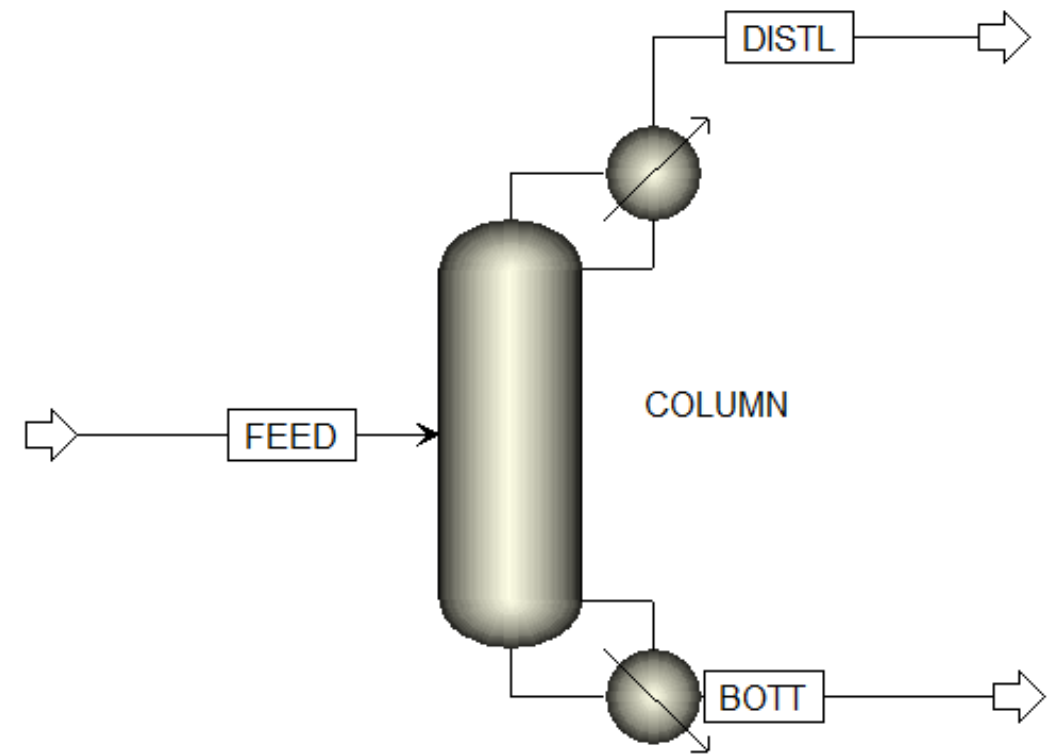


Example 3.3

Using Example 3.2, plot and report the effect of varying (i) reflux ratio and (ii) distillate rate to mole purity and recovery of methanol at the distillate stream. Use Sensitivity Analysis Tools. Using Column Design Specifications tools determine the reflux ratio and distillate rate required to achieve a separation of 99.9% recovery of methanol at distillate and 99.0% recovery of water at bottom.

Exercise 3.1

A distillation column is used to separate 1000 lbmol/hr of hydrocarbon mixtures at 190 F and 315 psia. The feed contain 0.26, 0.09, 0.25, 0.17, 0.11 and 0.12 mole fractions of methane (C1), ethane (C2), propane (C3), butane (nC4), pentane (nC5) and hexane (nC6), respectively. Using RKS-BM thermodynamic model, simulate the process and determine the reflux ratio to achieve 0.41 mole fraction of propane at distillate stream using design specifications tool.

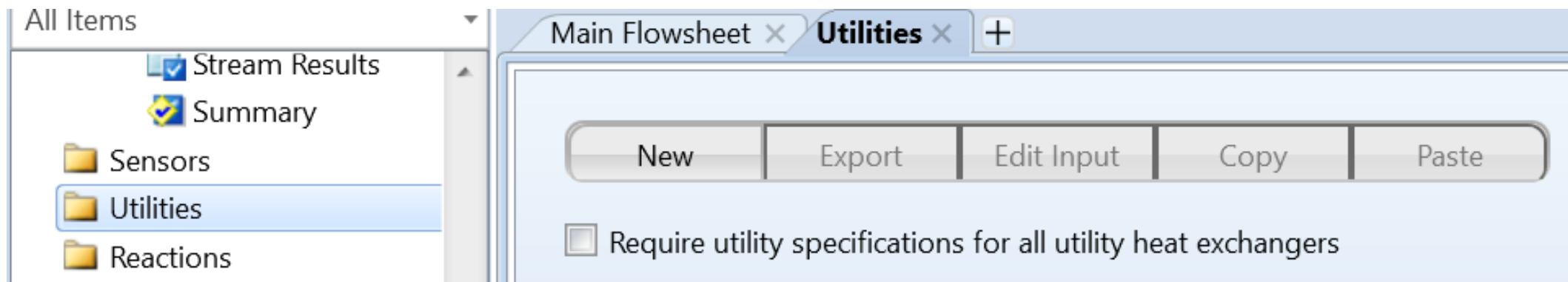


- Partial vapor condenser
- Kettle reboiler
- Number of stages: 15
- Feed stage: 8
- Distillate to feed ratio: 0.6
- Reflux ratio: XX
- Column pressure: 315 psia

Utilities Setup & Analysis

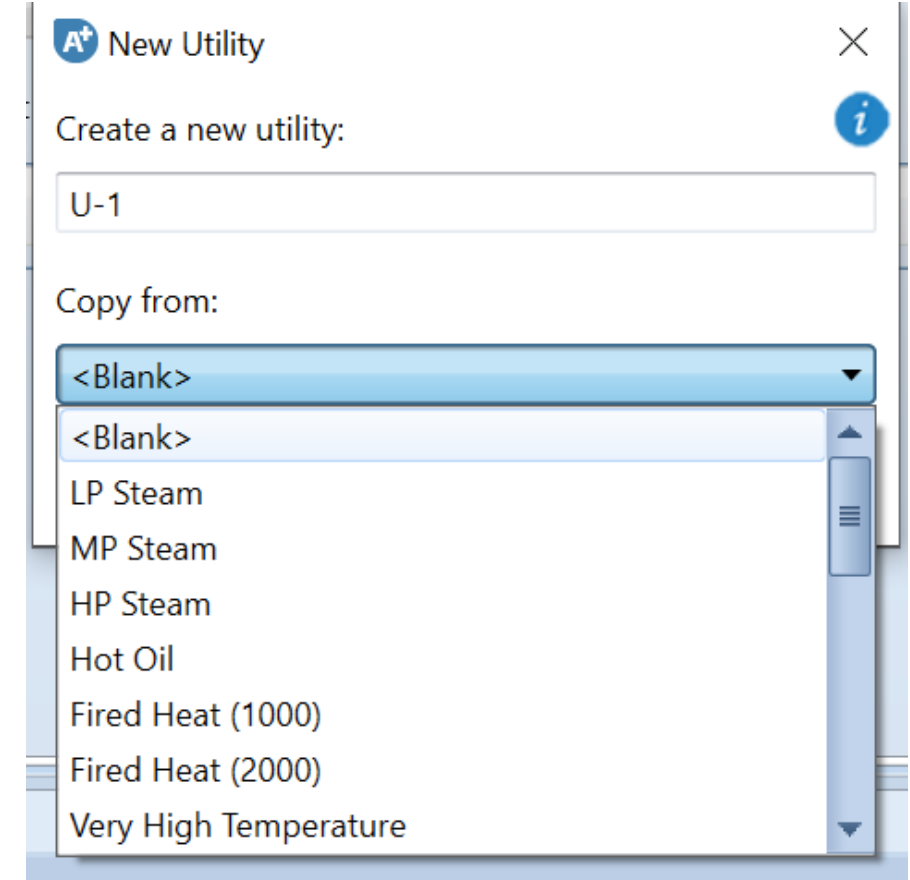
Utilities in Aspen Plus

- Allow calculation of: -
 - Individual unit operation energy/utility consumption
 - Energy costs,
 - Gas emissions
- Need to set up a utility ID using Utility form in Simulation pane.
- Implemented as variable utilities for each unit operation/block to computes its utilities usage.



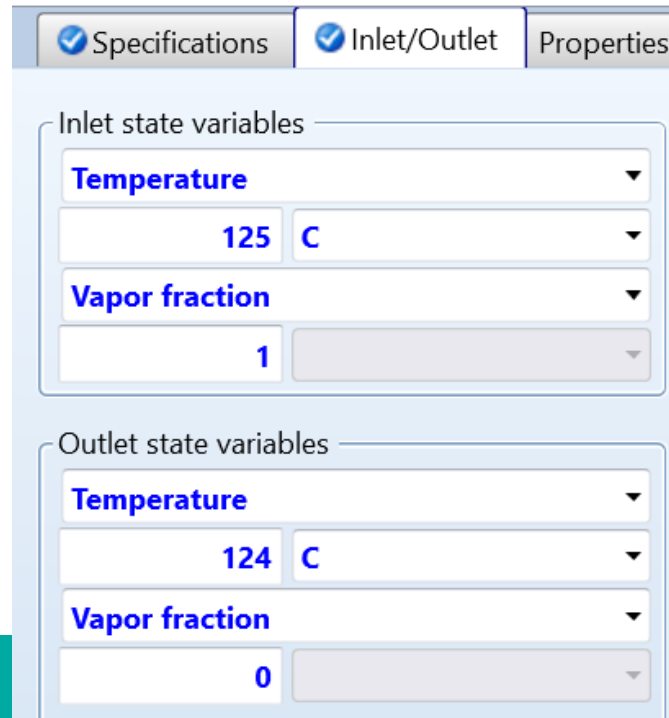
Utilities in Aspen Plus

- Go to Utilities to create New utility.
- The utilities available are coal, electricity, gas, oil, refrigerant, steam, water, and general.



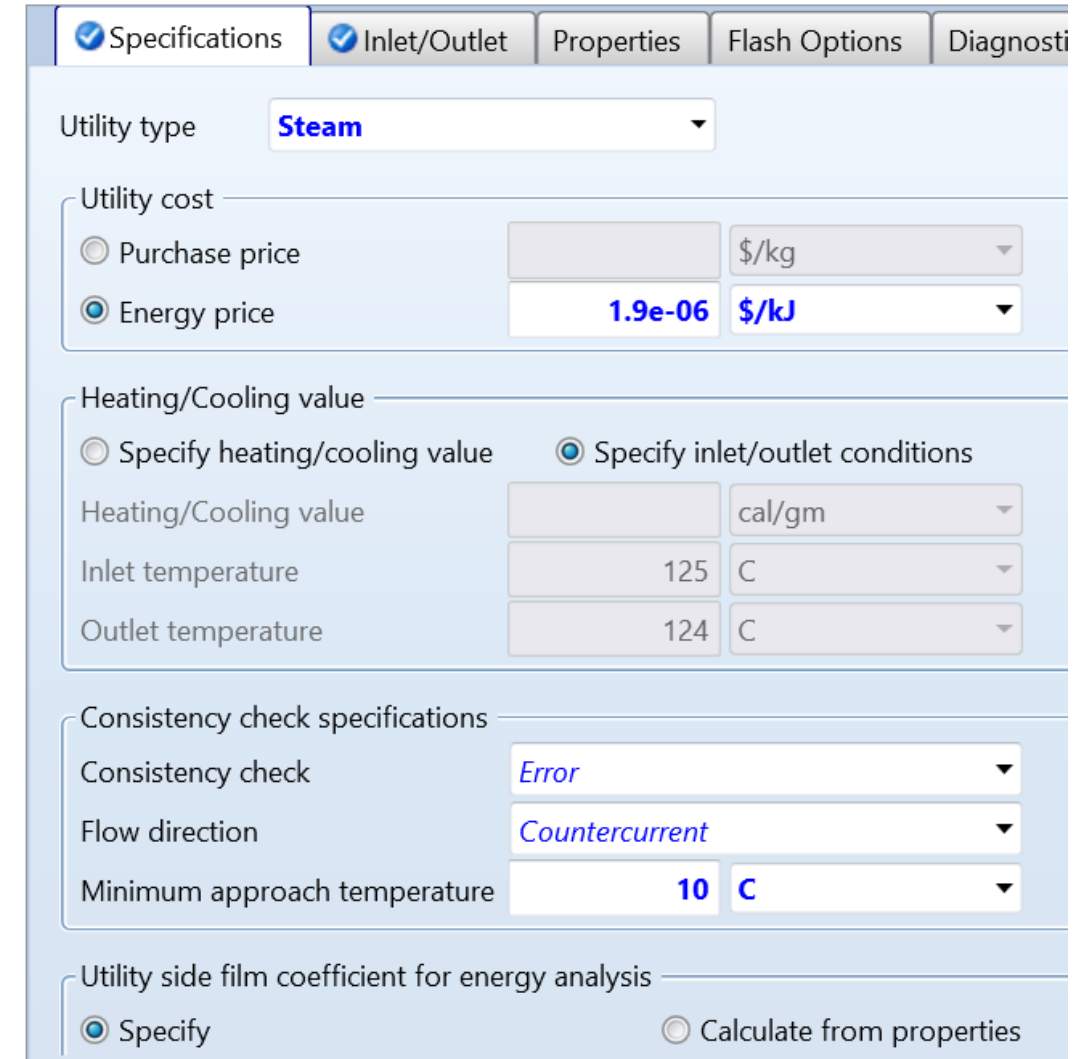
Utilities in Aspen Plus

- Specifications tab contain default information on energy price, heating/cooling value utility side film coefficient.
- Inlet/outlet tab contain the default utility state variable.



The image shows two tabs of the Aspen Plus Utilities dialog box. The 'Specifications' tab is active, showing inlet and outlet state variables. The 'Inlet/Outlet' tab is also visible, showing inlet and outlet conditions.

Tab	Variable	Value	Unit
Specifications	Inlet state variables	Temperature	125 C
	Vapor fraction	1	
	Outlet state variables	Temperature	124 C
	Vapor fraction	0	
Inlet/Outlet	Inlet state variables	Temperature	125 C
	Vapor fraction	1	

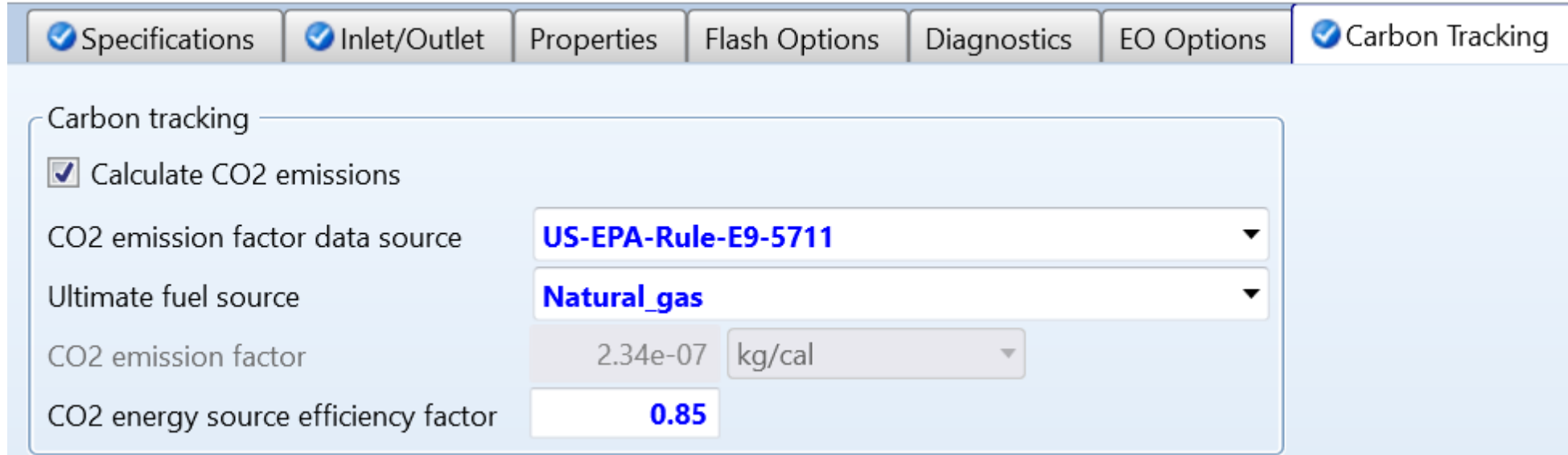


The image shows two tabs of the Aspen Plus Utilities dialog box. The 'Specifications' tab is active, showing utility type, cost, heating/cooling value, consistency check specifications, and utility side film coefficient. The 'Inlet/Outlet' tab is also visible, showing inlet and outlet conditions.

Tab	Variable	Value	Unit
Specifications	Utility type	Steam	
	Utility cost	Energy price	1.9e-06 \$/kJ
	Heating/Cooling value	Specify inlet/outlet conditions	
	Heating/Cooling value	cal/gm	
	Inlet temperature	125	C
	Outlet temperature	124	C
	Consistency check specifications	Consistency check	Error
	Flow direction	Countercurrent	
	Minimum approach temperature	10	C
	Utility side film coefficient for energy analysis	Specify	
Inlet/Outlet	Inlet state variables	Temperature	125 C
	Vapor fraction	1	

Utilities in Aspen Plus

- In the Carbon Tracking tab, Aspen Plus reports greenhouse gas emissions in terms of CO₂ equivalents of global warming potential (GWP).
- Available data source; IPCC's 2nd (SAR) and 4th (AR4) Assessment Reports, and the U.S. EPA's proposed rules from 2009.
- Variety of fuel source i.e., natural gas, coal, gasoline etc.

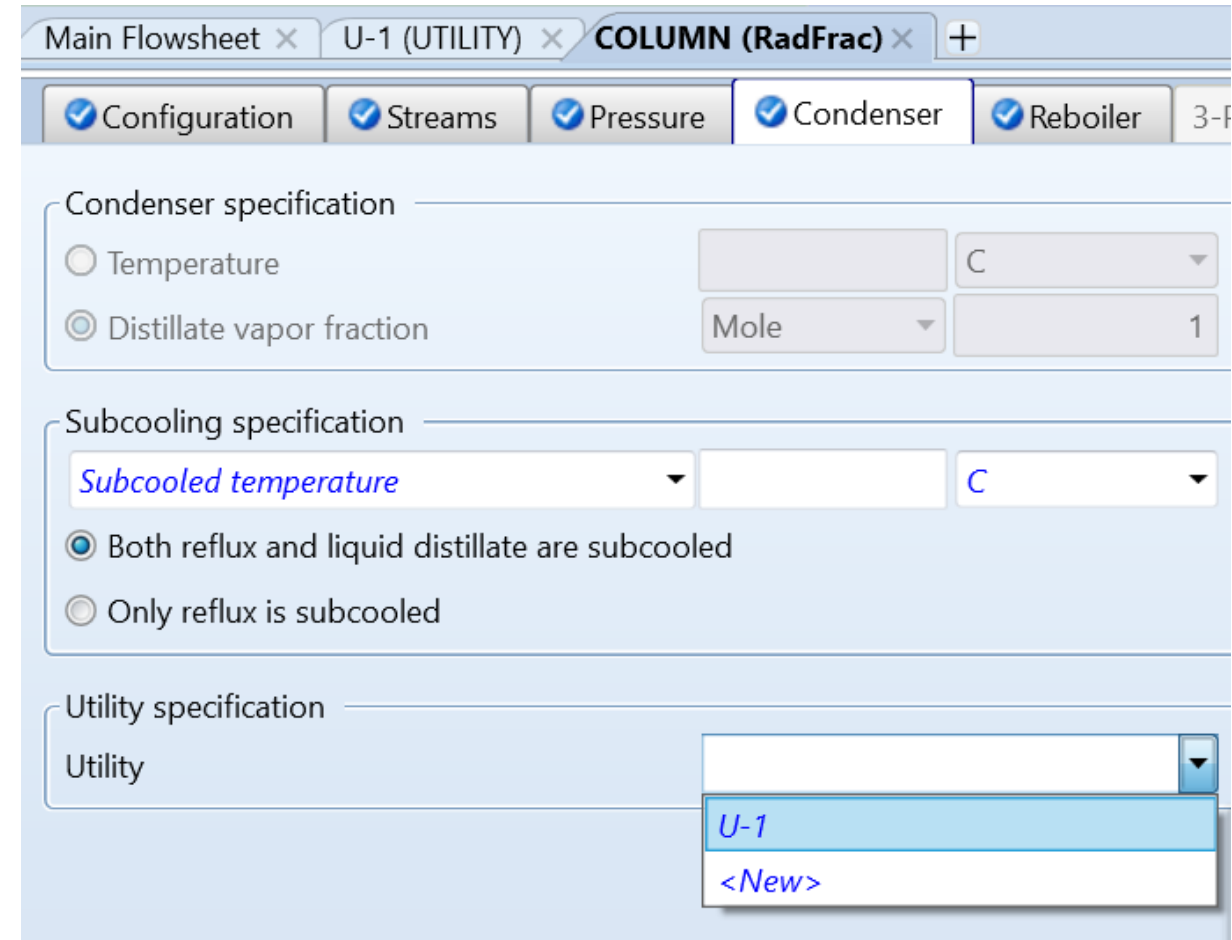


The screenshot shows the Aspen Plus Carbon Tracking tab. The 'Carbon Tracking' tab is selected, indicated by a blue checkmark. Below the tabs, the 'Carbon tracking' section is expanded. The 'Calculate CO2 emissions' checkbox is checked. The 'CO2 emission factor data source' is set to 'US-EPA-Rule-E9-5711'. The 'Ultimate fuel source' is set to 'Natural_gas'. The 'CO2 emission factor' is set to '2.34e-07' with a unit of 'kg/cal'. The 'CO2 energy source efficiency factor' is set to '0.85'.

Parameter	Value
Calculate CO2 emissions	☒
CO2 emission factor data source	US-EPA-Rule-E9-5711
Ultimate fuel source	Natural_gas
CO2 emission factor	2.34e-07 kg/cal
CO2 energy source efficiency factor	0.85

Utilities in Aspen Plus

- Utilities can be assigned to any block except MHeatX and Rplug.
- Assigning utilities to RadFrac block.



The screenshot displays the Aspen Plus configuration window for a RadFrac column. The window has tabs for 'Main Flowsheet', 'U-1 (UTILITY)', and 'COLUMN (RadFrac)'. The 'U-1 (UTILITY)' tab is active, showing the 'Configuration' sub-tab. The 'Condenser specification' section has 'Distillate vapor fraction' selected with a value of 1. The 'Subcooling specification' section has 'Subcooled temperature' selected with a value of C. The 'Utility specification' section shows a dropdown menu with 'U-1' selected and '<New>' as an option.

Exercise 3.2

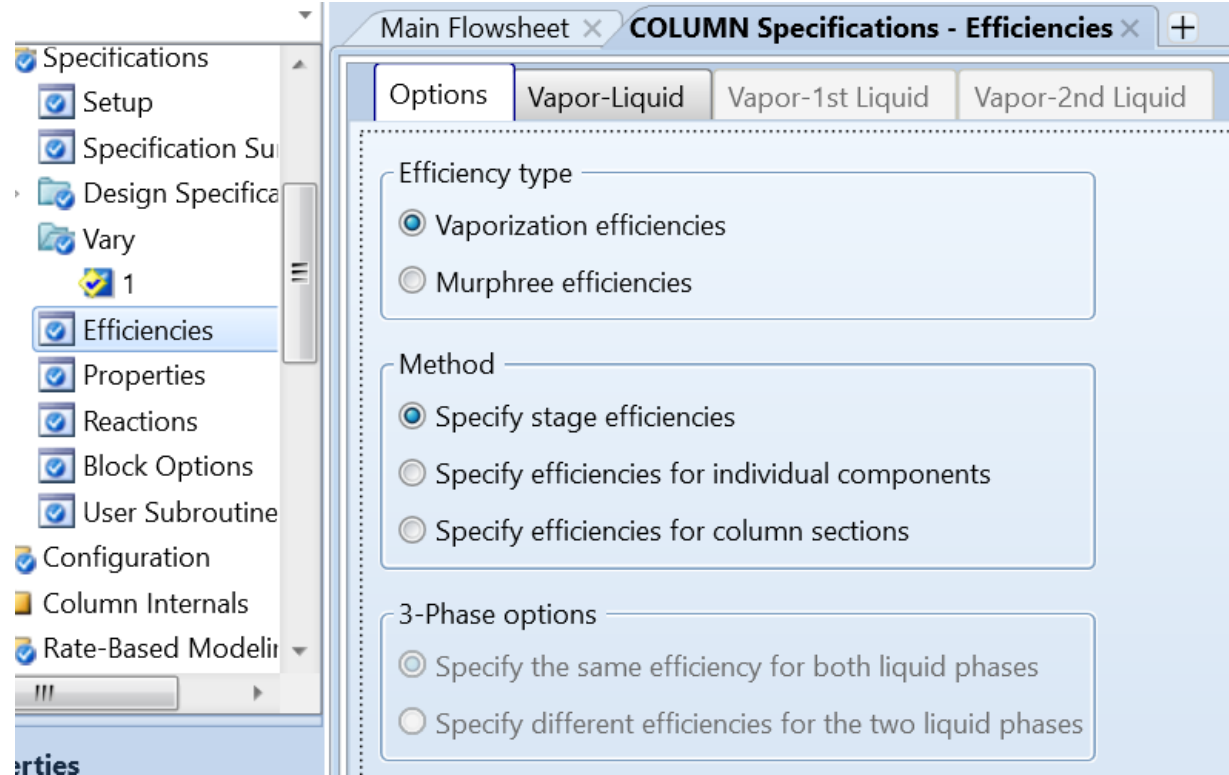
Using model developed in Exercise 3.1 setup a heating and cooling utilities using the following setup and report the results.

Parameter	Case 1		Case 2		Case 3	
	Reb	Cond	Reb	Cond	Reb	Cond
Type						
Flowrate						
Cost						
CO2						

Column Internals Setup & Analysis

Tray Efficiency

- $N_{stages} = N_{trays} + 2$ (cond & reb)
- Required to match real equipment and accurate column design.
- Murphee efficiencies are preferred as vaporization can alter the T profile and relative volatilities.
- Efficiencies can be a Vary in Design Specifications tool for model tuning.
- Specify stage efficiencies on Vapor-Liquid sheet and value not entered are linearly interpolated.
- Tray efficiency can be calculated using Rate-Based mode.



Example 3.4

Using Exercise 3.1, redesign the column number of stages for tray efficiency of 65%.
Run the model and report the new RR.

$$\text{Column trays} = 13 / 0.65 = 20$$

$$\text{Overall number of stages} = 20 + 2 = 22$$

$$\text{Feed tray} = (8 - 1) / 13 + 1 \approx 12$$

Example 3.4

1

Configuration Streams Pressure Condenser Reboiler 3-Phase Comm

Setup options

Calculation type *Equilibrium*

Number of stages **22** Stage Wizard

Condenser **Partial-Vapor**

Reboiler *Kettle*

Valid phases *Vapor-Liquid*

Convergence *Standard*

2

Configuration Streams Pressure Condenser Reboiler

Feed streams

Name	Stage	Convention
FEED	12	<i>Above-Stage</i>

*New RR = 2.58

3

Options Vapor-Liquid Vapor-1st Liquid Vapor

Efficiency type

☐ Vaporization efficiencies

☒ **Murphree efficiencies**

Method

☒ Specify stage efficiencies

☐ Specify efficiencies for individual components

☐ Specify efficiencies for column sections

3-Phase options

☐ Specify the same efficiency for both liquid phases

☐ Specify different efficiencies for the two liquid phases

4

Options Vapor-Liquid Vapor

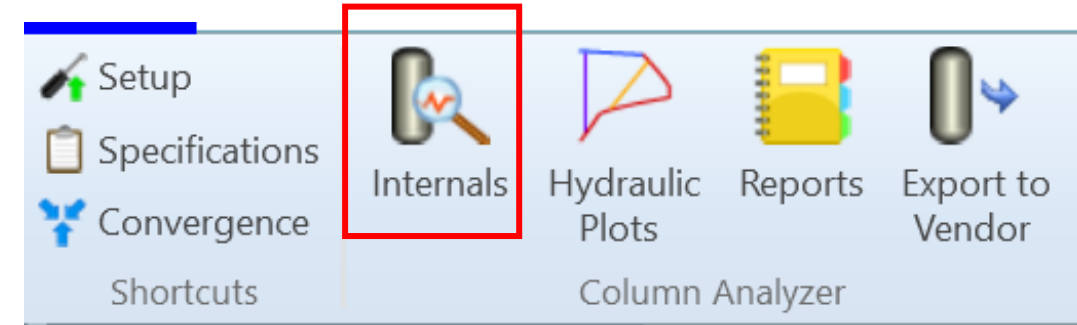
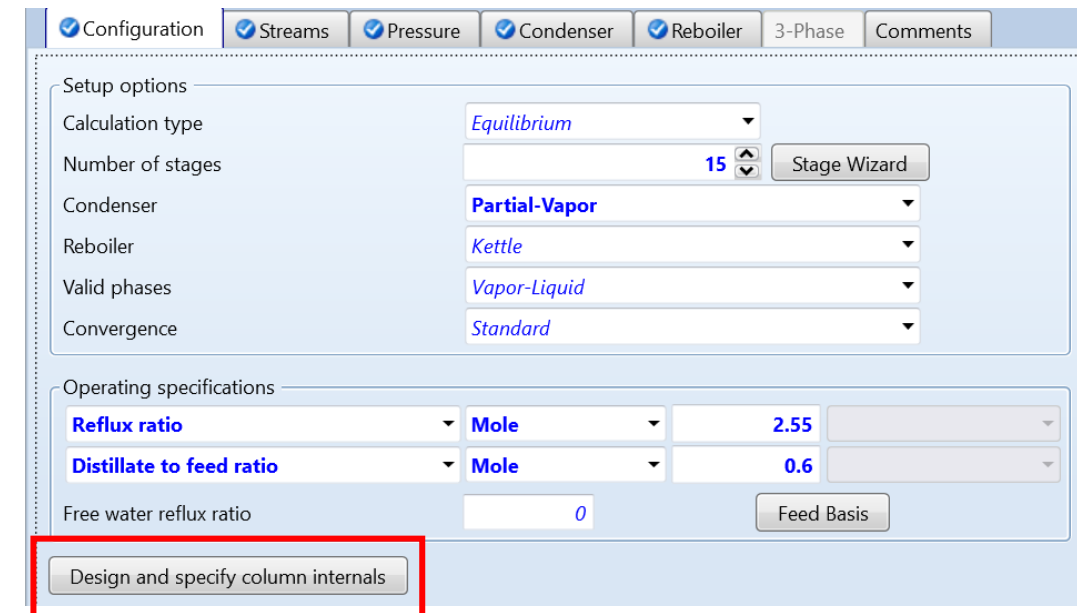
Stage efficiency

Murphree

Stage	Efficiency
1	1
2	0.65
21	0.65
22	1

Column Sizing & Rating

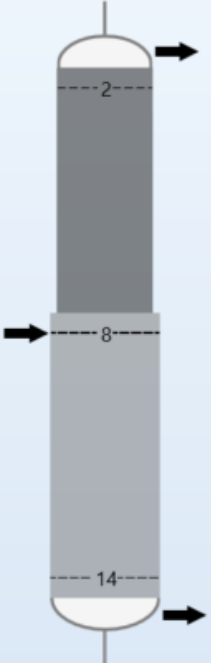
- Aspen Plus calculates sizing and performance parameters for trayed and packed columns such as:
 - Column diameter
 - Flooding approach or approach to maximum capacity
 - Downcomer backup
 - Pressure drop
- These parameters are based on:
 - Column loadings
 - Transport properties
 - Tray geometry
 - Packing characteristics
- Available in RadFrac, MultiFrac and PetroFrac model blocks.
- 5 five commonly-used tray types i.e. Bubble caps, Sieve, Glitsch Ballast®, Koch Flexitray® and Nutter Float Valve.
- Structured packings include Goodloe®, Glitsch Grid®, Norton Intalox, Sulzer BX, CY, Mellapak, and Kerapak, Koch Flexipac, Flexeramic, Flexigrid, Raschig Super-Pak and Ralu-Pak.

Column Sizing & Rating

Sections

Status Active



View Hydraulic Plots

Column description

Input Complete

Add New

Auto Section

Duplicate

Import Template

Export Template

View Internals Summary

Name	Start Stage	End Stage	Mode	Internal Type	Tray/Packing Type	Tray Details		Packing Details			Tray Spacing/Section Packed Height		Diameter		Details	
						Number of Passes	Number of Downcomers	Vendor	Material	Dimension						
CS-1	2	7	Interactive sizing	Trayed	SIEVE	1					0.6096	meter	1.61876	meter	View	✗
CS-2	8	14	Interactive sizing	Trayed	SIEVE	1					0.6096	meter	1.87996	meter	View	✗

☒ Don't update pressure drop
 ☐ Update pressure drop from top stage
 ☐ Update pressure drop from bottom stage

☒ Include static vapor head in pressure drop calculations
 ☐ Calculate pressure drop across sump

Sump

Diameter

1.87996

meter

☒ Liquid residence time

0.0166667

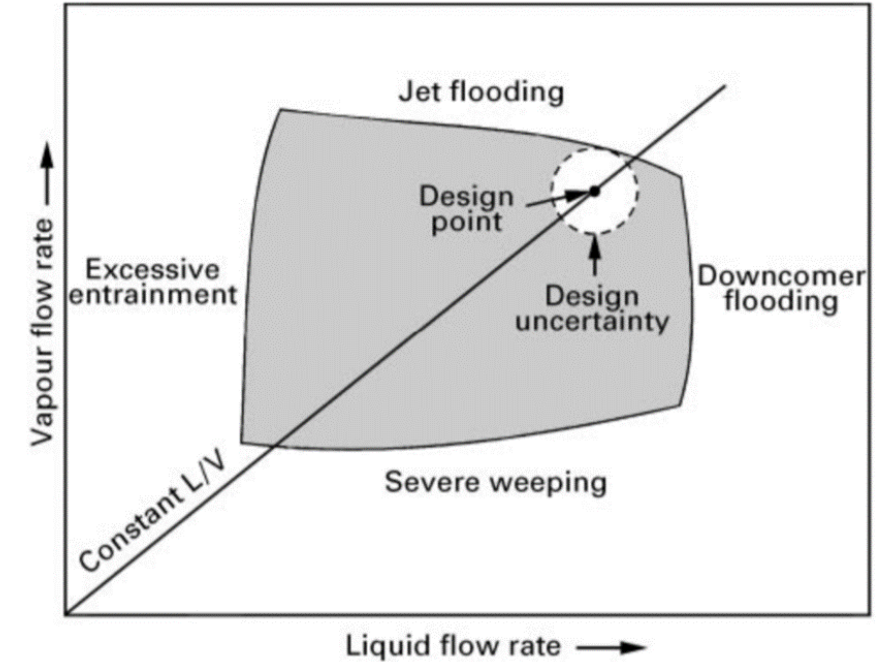
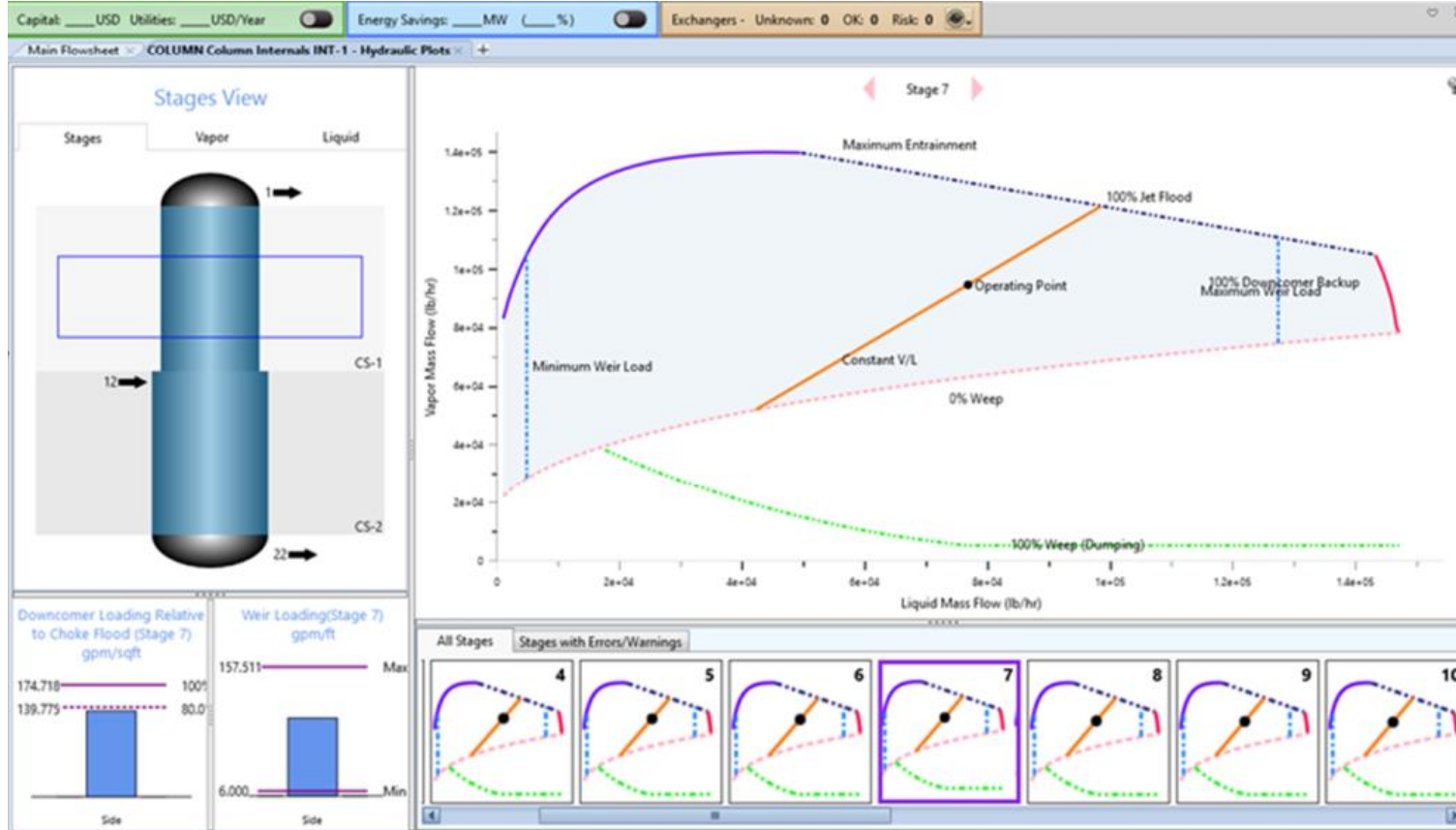
hr

☐ Liquid level

meter

- The column can be divided sections with different tray type, packing type, and diameter.
- The tray details can vary from section to section with unlimited number of sections.
- Auto-sectioning of column based on feed draw locations or internal flows.
- The calculations are based on vendor-recommended procedures whenever these are available.
- When vendor procedures are not available, well-established literature methods are used.
- Design advice to mitigate observed hydraulic issues.
- Column calculations do not affect column separation (supplementary calculations).

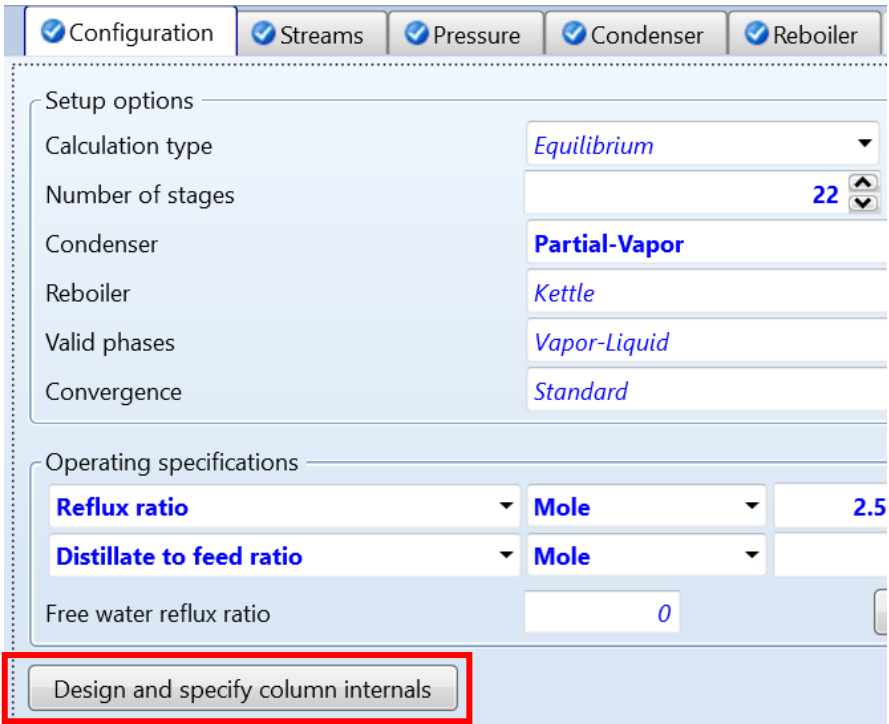
Column Sizing & Rating



Example 3.5

Using Exercise 3.4, deactivate design specification and vary option. Create a tray column and perform column sizing and analysis.

1



Configuration Streams Pressure Condenser Reboiler

Setup options

Calculation type: *Equilibrium*

Number of stages: 22

Condenser: *Partial-Vapor*

Reboiler: *Kettle*

Valid phases: *Vapor-Liquid*

Convergence: *Standard*

Operating specifications

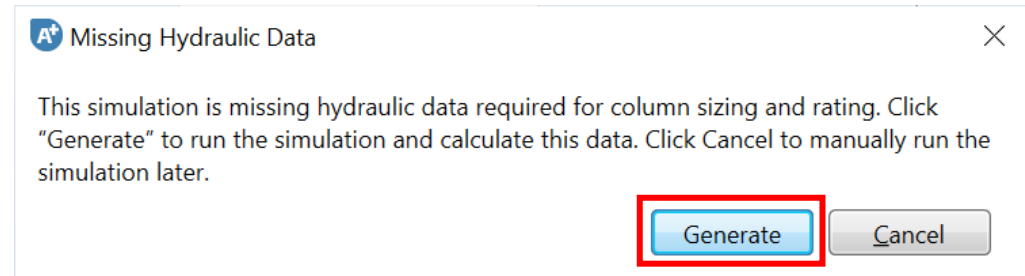
Reflux ratio: *Mole* 2.5

Distillate to feed ratio: *Mole*

Free water reflux ratio: 0

Design and specify column internals

2



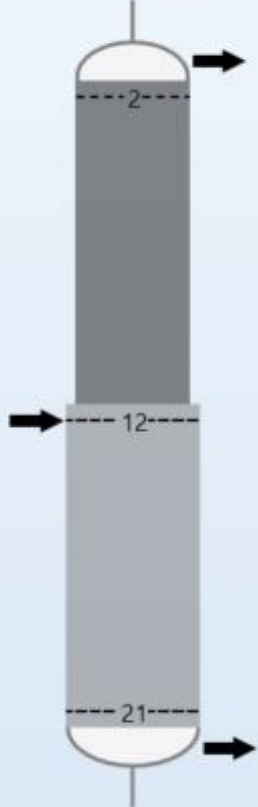
Missing Hydraulic Data

This simulation is missing hydraulic data required for column sizing and rating. Click "Generate" to run the simulation and calculate this data. Click Cancel to manually run the simulation later.

Generate Cancel

- ☒ Reactions
- ☒ Block Options
- ☒ User Subroutine
- ▶ ☒ Configuration
- ▶ ☒ Column Internals
 - ▶ ☒ INT-1
 - ▶ ☒ Sections
 - ☒ Hydraulic Plots
 - ☒ Column Hydraulic
- ▶ ☒ Rate-Based Modeling
- ▶ ☒ Analysis
- ▶ ☒ Convergence
- ▶ ☒ Dynamics
- ▶ ☒ EO Modeling
- ☒ Results
- ☒ Profiles
- ☒ Stream Results
- ☒ Summary
- ▶ Sensors
- ▶ Utilities
- ▶ Reactions
- ▶ Convergence
- ▶ Flowsheeting Options
- ▶ Model Analysis Tools
- ▶ EO Configuration

Status: Active



Column description

3

Add New

Auto Section ▼

Duplicate
Import Template
Export Template
View Internals Summary

	Name	Start Stage	End Stage	Mode	Internal Type	Tray/Packing Type	Tray Details		
							Number of Passes	Number of Downcomers	Vendor
▶	CS-1	2	11	Interactive sizing	Trayed	SIEVE	1		
▶	CS-2	12	21	Interactive sizing	Trayed	SIEVE	1		

☒ Don't update pressure drop

☐ Update pressure drop from top stage

☐ Update pressure drop from bottom stage

☒ Include static vapor head in pressure drop calculations

☐ Calculate pressure drop across sump

Sump

Diameter

1.76235 meter ▼

☒ Liquid residence time

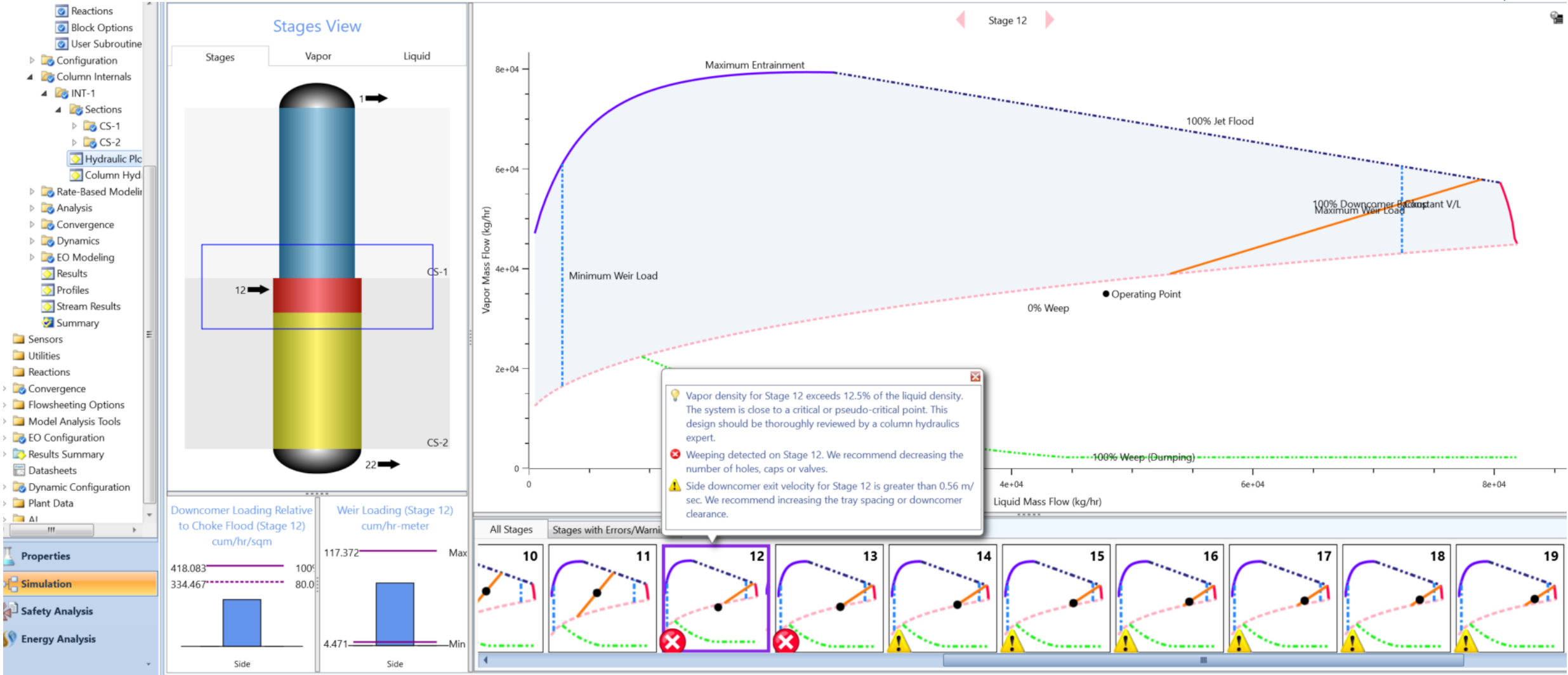
0.0166667 hr ▼

☐ Liquid level

meter ▼

View Hydraulic Plots

4



☒ Reactions
☒ Block Options
☒ User Subroutine
 Configuration
 Column Internals
 INT-1
 Sections
 CS-1
 CS-2
 Geometry
 Results
 Hydraulic Plot
 Column Hydraulic

Properties
Simulation
Safety Analysis
Energy Analysis

☒ Geometry

☐ Design Parameters

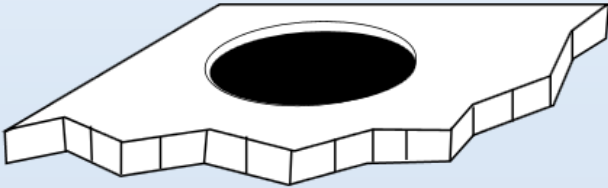
☐ Tray Geometry Summary

Name Start stage End stage Status Mode ☒ Interactive sizing ☐ Rating

Section type ☒ Trayed ☐ Packed

Tray type Downcomer arrangement Number of passes

Hole Diameter

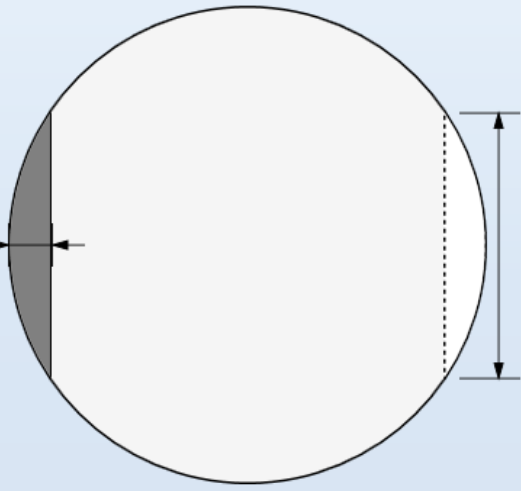


☒ Hole area / Active area
☐ Number of holes
 Deck thickness

Side Downcomer Width

Top

Bottom



Side Weir Length

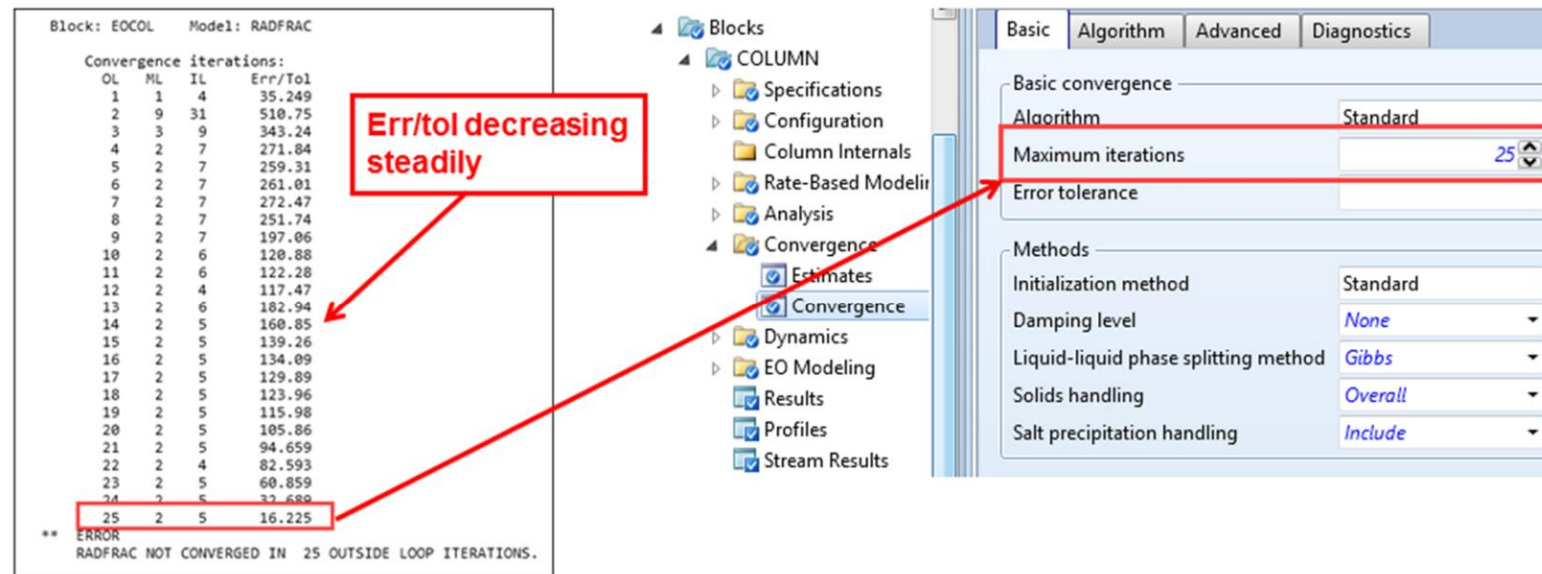
Diameter

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RadFrac Convergence Tips

- Ensure that column operating conditions are feasible.
- Good practice to reset simulation after each changes before Run.
- Check physical properties issues (property method, parameter availability etc.) are properly addressed.
- If the column err/tol is decreasing consistently to 1, increase the maximum iterations. Default is 25.



RadFrac Convergence Tips

- Converge without design specification and vary option to initialize the column before activate the option.
- Provide temperature (useful for absorbers) and/or composition (useful for highly non-ideal system) estimates.
- Consider different convergence methods. Note, Standard method works well for most cases.

Exercise 3.3

A feed stream with 1000 kg/h containing mixture of 30 wt.% benzene, 40wt % toluene and the rest ethylbenzene need to be separated into its pure component using two distillation columns in series. Note that bottom stream of the first column serves as the feed to the second column. It is known that the feed temperature is 30 C at 1 atm. Use the following information to simulate the process.

Distillation 1	Distillation 2
Number of stages = 14	Number of stages = 25
Feed stage = 6 (Above stage)	Feed stage = 12 (Above stage)
Molar reflux ratio = 2	Molar reflux ratio = 2
Distillate rate = 297 kg/h	Distillate rate = 392 kg/h
Column pressure = 0.5 atm	Column pressure = 0.5 atm
Total condenser & Kettle reboiler	Total condenser & Kettle reboiler

Exercise 3.3

1. Determine suitable thermodynamic model to be used for the process. Compare with NIST data.
2. Simulate the process and report the following results,

Benzene distillate mass fraction : _____

Toluene distillate mass fraction : _____

1st column cond / reb duty [kW] : _____ / _____

2nd column cond / reb duty [kW] : _____ / _____

Exercise 3.3

3. Set up a design specification for each column to achieve 99 wt.% benzene in the distillate of 1st column and 98 wt.% toluene in the distillate of 2nd using column design specifications tool. Report the following results,

1st column vary variable / value : _____ / _____

2nd column vary variable / value : _____ / _____

1st column cond / reb duty [kW] : _____ / _____

2nd column cond / reb duty [kW] : _____ / _____

Exercise 3.3

4. If tray efficiency for each column is assumed 65%, deactivate design specifications, simulate the process and report the following results,

Benzene distillate mass fraction : _____

Toluene distillate mass fraction : _____

1st column cond / reb duty [kW] : _____ / _____

2nd column cond / reb duty [kW] : _____ / _____